Can Field Data Catch a Safety Regression?

Test coverage, sequential monitoring and automated risk reporting for automated-driving software releases

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Summary

Automated driving systems (ADS) receive frequent software updates. After each release, a safety team needs to know two things: whether testing before the release was enough, and how quickly field data would reveal a release that made driving less safe. This note answers both questions quantitatively. It uses event rates per mile derived from public NHTSA crash reports for Waymo's rider-only fleet (271.3 million miles through June 2026; companion analysis, DOI 10.5281/zenodo.23732384) and standard statistical methods.

Findings:

- **Field data catches only frequent outcomes quickly.** With the whole fleet driving about 2.9 million miles a week, a 25% increase in the all-crash rate is detectable in about **6 weeks**. The same increase in any-injury crashes takes about **65 weeks**. For serious-injury-or-worse crashes it takes about **1,730 weeks (33 years)**.
- **So release gating for severe outcomes must rely on testing and leading indicators.** A frequent leading indicator (assumed at 100 events per million miles) would reveal a 25% regression within about three days of fleet driving.
- **Scenario testing needs hundreds to thousands of runs per scenario type.** Showing a failure probability below 1% at 95% confidence takes 299 failure-free runs; below 0.1% takes 2,995.
- **Sequential monitoring works, but false alarms add up.** Poisson CUSUM charts tuned to one false alarm per metric every five years still give an **85%** chance of at least one false alarm somewhere across five metrics in two years. Every alarm needs a fast, cheap triage step.

- **The reporting can be automated.** A scored risk register following the ISO 31000 lifecycle and a one-page release health report are generated from the data by one script.

1. The question

A risk-management team for an ADS has to say what test coverage and field monitoring are needed to protect against regressions: a new release that performs worse than the one before. Two questions follow:

1. **Field monitoring:** after a release goes out, how many miles of driving does it take to detect a given increase in an event rate, and how should alarms be set?
2. **Test coverage:** if field data cannot catch a regression in time, how much pre-release testing is needed instead?

2. Data and assumptions

Table 1. Event rates used (per million miles). PUBLISHED-DERIVED values come from public NHTSA Standing General Order crash reports for Waymo's rider-only fleet, analyzed in the companion repository; ASSUMED values are illustrative.

Metric	Rate per million miles	Basis
All reported crashes	7.13	Published-derived: 1,934 crashed vehicles in 271.3M miles
Any-injury crashes	0.612	Published-derived: posterior median, 166 events
Airbag deployment	0.266	Published-derived: posterior median, 72 events
Serious injury or worse	0.023	Published-derived: posterior median, 6 events
Leading indicator (e.g. hard braking, near misses)	100	Assumed; not public

Fleet exposure. About 12.7 million miles per month (2.9 million per week), derived from the 542 crashes reported in January–June 2026 at 7.13 crashes per million miles. Every release is assumed to run on the whole fleet.

What is not used. No Waymo internal data. Simulated release histories are synthetic; the risk register's operational hazards (H6–H8) have no public data and are flagged as data gaps.

3. Methods

Detectability (exact Poisson test). The baseline rate λ_0 is known from long fleet history. After a release, N events are observed in M million miles. An alarm is raised if $P(N \geq n \mid \lambda_0 M) \leq \alpha = 0.05$. The power to detect a regression to $\lambda_1 = (1 + r)\lambda_0$ is

$$\text{power}\,(M) = P(N \geq c \mid \lambda_1 M), \qquad c = \min\{n : P(N \geq n \mid \lambda_0 M) \leq \alpha\}$$

and the exposure reported is the smallest M after which power stays at or above 80%. A release-versus-release comparison, with no known baseline, needs roughly twice as many miles (normal approximation on the log rate ratio; `results/table1_detectability.csv`).

Test coverage (zero-failure demonstration). To show that a scenario type's failure probability is below p at confidence C with n failure-free runs:

$$n = \left\lceil \frac{\ln\,(1-C)}{\ln\,(1-p)} \right\rceil \approx \frac{3}{p} \qquad (C = 0.95)$$

Sequential monitoring (Poisson CUSUM; Lucas, 1985). With weekly counts X_t, in-control mean m_0 and the regression to catch $m_1 = 1.25\,m_0$:

$$k = \frac{m_1 - m_0}{\ln\,(m_1/m_0)}, \qquad S_t = \max\,(0,\ S_{t-1} + X_t - k)\,, \qquad \text{alarm if } S_t \geq h$$

h is set by simulation so that the mean time between false alarms (ARL_0) is 260 weeks per metric. After each alarm the chart restarts, as it would after an investigation.

Bayesian check. With a Jeffreys prior, the posterior rate after N events in M million miles is Gamma(N + ½, M), giving $P(\lambda > \lambda_0 \mid \text{data})$.

Risk register. Hazards are scored by their injury-crash rate per 100 million miles, with Gamma posteriors for counts and Beta posteriors for P(injury | crash), following the ISO 31000 sequence: identify, analyze, evaluate, treat, monitor and report.

4. Results

4.1 How long field data needs to catch a regression

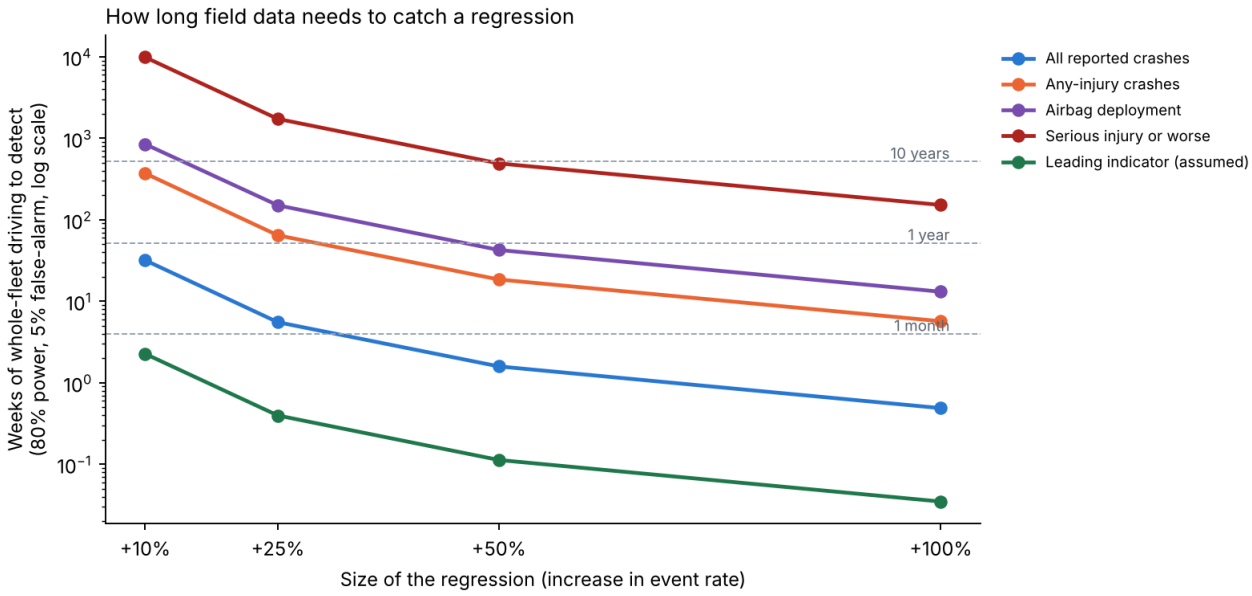


Figure 1. Weeks of whole-fleet driving needed to detect a regression with 80% power and a 5% false-alarm rate, by metric and size of regression (log scale).

Table 2. Weeks of whole-fleet driving to detect a regression (80% power, 5% false alarm, known baseline).

Metric	+10%	+25%	+50%	+100%
Leading indicator (assumed)	2.3	0.4	0.1	<0.1
All reported crashes	31.9	5.5	1.6	0.5
Any-injury crashes	371	64.5	18.5	5.7
Airbag deployment	846	150	42.6	13.1
Serious injury or worse	9,859	1,728	492	152

A release typically lives for weeks. Field data can therefore protect against regressions in frequent outcomes, but not in severe ones: even a doubling of the serious-injury rate would take about three years of whole-fleet driving to show.

4.2 How much testing replaces the missing field evidence

Table 3. Failure-free scenario runs needed to show a failure probability below a target.

Target failure probability	90% confidence	95% confidence	99% confidence
10%	22	29	44
1%	230	299	459
0.1%	2,302	2,995	4,603
0.01%	23,025	29,956	46,050

Simulation makes these numbers feasible; its credibility (validation against the field) then becomes the key evidence.

4.3 Sequential monitoring in practice

Table 4. Poisson CUSUM design per metric (weekly counts; tuned for +25%; ARL_0 about 260 weeks). Median weeks to the first alarm after a regression starts.

Metric	Expected weekly count	h	+25%	+50%	+100%
All reported crashes	20.8	16.9	6	3	1
Any-injury crashes	1.79	10.9	30	14	7
Airbag deployment	0.78	8.5	45	24	12
Serious injury or worse	0.067	3.3	105	70	43
Leading indicator (assumed)	292	11.4	1	1	1

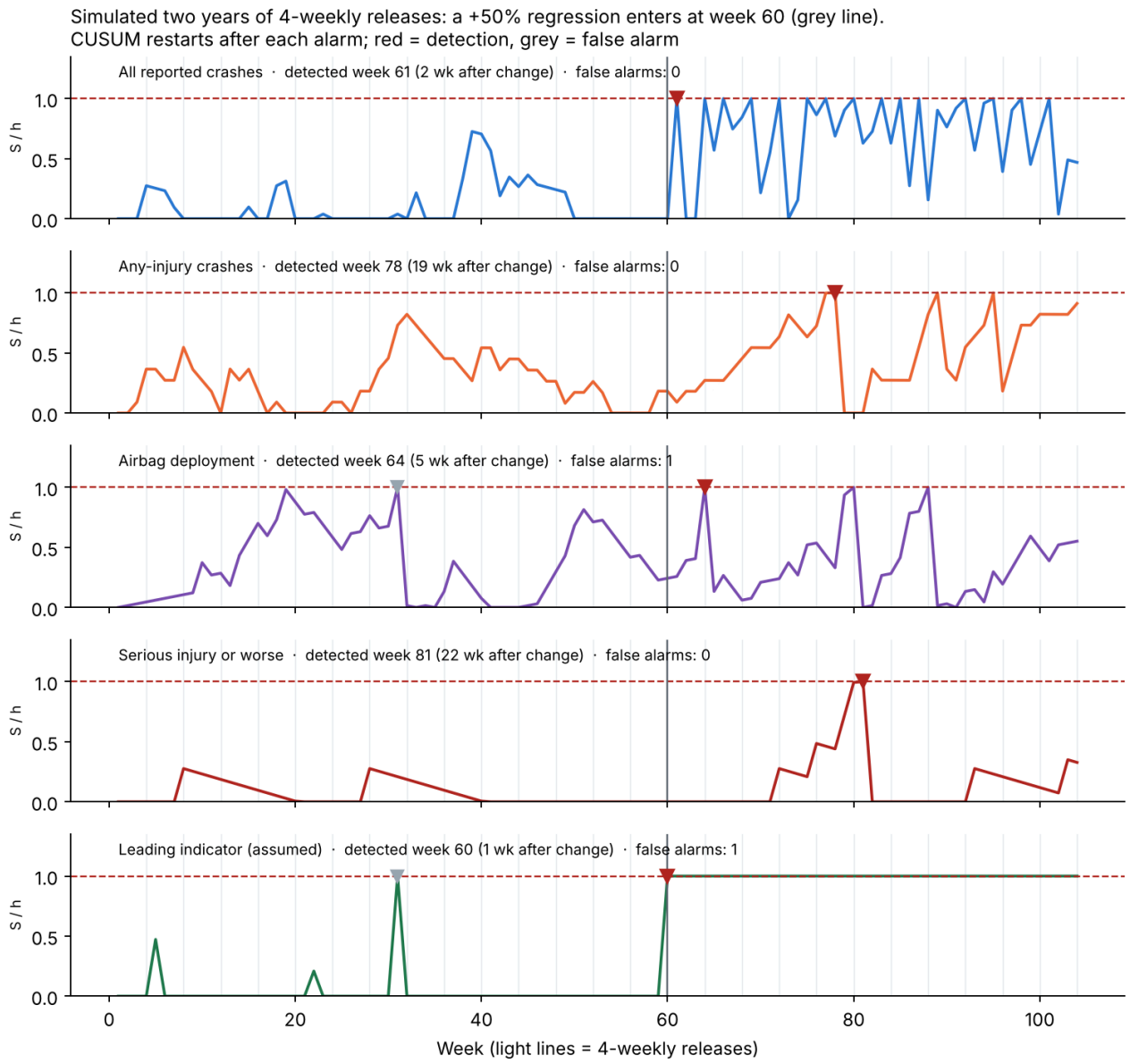


Figure 2. One simulated two-year history with 4-weekly releases and a +50% regression in every metric from week 60. Red markers: detections; grey markers: false alarms.

In the simulated history (Figure 2), the leading indicator alarmed in the first week of the regression and all crashes in the second. Airbag deployments took 5 weeks, any-injury crashes 19 and serious injuries 22; the serious-injury metric was lucky here, since its median is 70 weeks. Two false alarms occurred before the change. Across five metrics each tuned to one false alarm per five years, the chance of at least one false alarm in two years is 85%.

4.4 Risk register and automated reporting

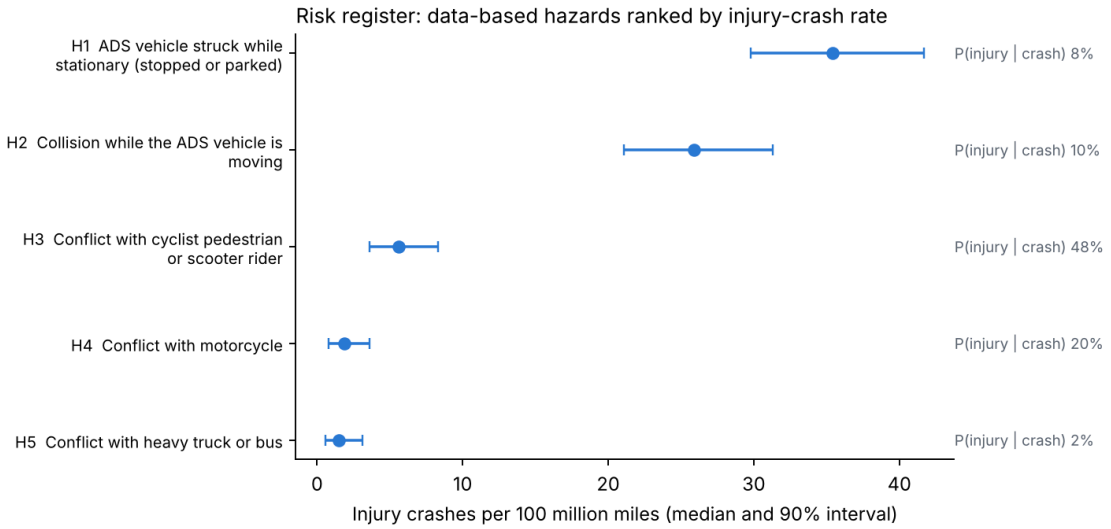


Figure 3. Data-based hazards ranked by injury-crash rate per 100 million miles (median and 90% interval).

Table 5. Scored risk register (ISO 31000 lifecycle). H1–H5 are data-based (companion analysis); H6–H8 have no public data and are flagged as data gaps.

ID	Hazard	Crash rate per M mi	P(injury given crash)	Injury crashes per 100M mi (90% interval)	Treatment status
H1	Struck while stationary	4.40	8.1%	35.4 (29.8–41.7)	Monitor
H2	Collision while moving	2.73	9.5%	25.9 (21.1–31.3)	Monitor
H3	Cyclist, pedestrian or scooter rider	0.11	48.4%	5.6 (3.6–8.3)	Reduce
H4	Motorcycle	0.09	20.4%	1.9 (0.8–3.6)	Reduce
H5	Heavy truck or bus	0.83	1.9%	1.5 (0.6–3.1)	Accept and monitor
H6	Sensor or compute degradation, in-lane stop	–	–	data gap	Data gap
H7	Delayed or incorrect remote assistance	–	–	data gap	Data gap
H8	Behavior regression from a release	–	–	monitored (Section 4.3)	Monitor

The script also writes a one-page release health report (`results/release_health_report.html`): bottom line first, monitoring status per metric, what field data can and cannot catch, top risks, data gaps and recommended actions. Every number in it is computed, so it can be regenerated for each release without manual editing.

Release health report — simulated fleet, week 104

Generated automatically by scripts/run_all.py. Simulated data on baseline rates from public NHTSA SGO reports (Repo 1); not Waymo internal data.

> **Bottom line.** 5 of 5 monitored metrics alarmed after a simulated +50% regression entered at week 60: All reported crashes, Any-injury crashes, Airbag deployment, Serious injury or worse, Leading indicator (assumed). Slowest to alarm: Serious injury or worse (22 weeks). Recommend holding the next release and bisecting the week-60 release.

1. Monitoring status (Poisson CUSUM tuned for +25%, one false alarm per metric every 5 years on average)

Metric	Status	First alarm after week 60	False alarms before	Events since week 60 vs expected	P(rate increased)
All reported crashes	RED	61	0	1409 vs 938.1	1.000
Any-injury crashes	RED	78	0	134 vs 80.5	1.000
Airbag deployment	RED	64	1	67 vs 35.0	1.000
Serious injury or worse	RED	81	0	7 vs 3.0	0.979
Leading indicator (assumed)	RED	60	1	19610 vs 13159.3	1.000

With five metrics each tuned to one false alarm per five years, the chance of at least one false alarm somewhere in two years is 85%: every alarm needs a fast, cheap triage step.

2. What field data can and cannot catch

Weeks of whole-fleet driving (2.9M miles/week) to detect a +25% regression with 80% power:

Metric	Weeks
All reported crashes	5.5
Any-injury crashes	64.5
Airbag deployment	150.4
Serious injury or worse	1,728.4
Leading indicator (assumed)	0.4

Serious-injury regressions cannot be caught from field outcomes on any release timescale: release gating must rely on scenario testing and leading indicators.

3. Top data-based risks (injury crashes per 100M miles, 90% interval)

ID	Hazard	Rate	Treatment status
H1	ADS vehicle struck while stationary (stopped or parked)	35.4 (29.8–41.7)	Monitor
H2	Collision while the ADS vehicle is moving	25.9 (21.1–31.3)	Monitor
H3	Conflict with cyclist pedestrian or scooter rider	5.6 (3.6–8.3)	Reduce

4. Data gaps to close

- H6: Sensor or compute degradation leading to an in-lane stop — owner Hardware / Systems
- H7: Delayed or incorrect remote-assistance guidance — owner Operations

5. Recommended actions

1. Hold the next release; bisect the week-60 release with the scenario library.
2. Add the alarmed scenario types to the regression test set.
3. Instrument leading indicators for H6–H8 so they can be monitored.

Figure 4. The automatically generated release health report for the simulated history.

5. What this means for release gating

1. **Gate on testing for severe outcomes.** Field outcomes cannot protect a single release against serious-injury regressions. A risk-weighted scenario library, re-run on every release, must carry that load, with its credibility established against field data.
2. **Monitor leading indicators.** Frequent precursors that correlate with severe outcomes give field monitoring real power. Establishing that correlation is itself a research task.
3. **Design alarms as a system.** Tune each chart to the regression size that matters and budget the family-wise false-alarm rate, with a fast triage step for each alarm.
4. **Automate the reporting.** A computed risk register and release report keep assessments consistent as the product scope grows.

6. Limitations

- Rates are from public, self-reported crash data, with the limitations described in the companion analysis.
- Every release is assumed to run on the whole fleet at a constant mileage; staged rollouts would lengthen detection times.
- Weekly counts are assumed Poisson; overdispersion (for example from weather or city mix) would need negative-binomial charts.
- The leading-indicator rate and its link to severe outcomes are assumed.
- The risk register is illustrative and does not describe any company's actual register.

7. Reproduce

```
pip install -r requirements.txt
python tests/test_core.py        # 8 tests
python scripts/run_all.py        # all tables, figures and the release health report (about
30 s)
```

References

- Albati, M. (2026). How much evidence is enough? A Bayesian rare-event analysis of robotaxi crash data. Zenodo. doi:10.5281/zenodo.23732384
- International Organization for Standardization (2018). *ISO 31000:2018 Risk management: Guidelines*.
- International Organization for Standardization (2019). *IEC 31010:2019 Risk management: Risk assessment techniques*.
- Kalra, N., Paddock, S. M. (2016). *Driving to Safety: How Many Miles of Driving Would It Take to Demonstrate Autonomous Vehicle Reliability?* RAND RR-1478-RC.
- Lucas, J. M. (1985). Counted data CUSUM's. *Technometrics* 27(2), 129–144.
- Montgomery, D. C. (2019). *Introduction to Statistical Quality Control*, 8th ed. Wiley.
- National Highway Traffic Safety Administration. Standing General Order 2021-01 on crash reporting: ADS incident reports.